

A mod-8 period-4 theorem and two companion
identities for recurrence-defined tatami counting
sequences
(Lean 4 machine-checked claim of record)

Aurora0134*

September 27, 2026

Abstract

We record three theorems about two recurrence-defined integer sequences: $a(n)$, given by the base values and the order-6 recurrence of OEIS entry A180970, and $b(n)$, a corner-defect companion sequence satisfying the same order-6 recurrence. For all $n \geq 4$, $a(n) \bmod 8$ depends only on $n \bmod 4$, with pattern $(4, 2, 4, 6)$; $b(n)$ is odd for all $n \geq 4$; and $20b(n)$ equals a fixed linear combination of $a(n), \dots, a(n-5)$ for all $n \geq 10$. All proofs are machine-checked by the Lean 4 kernel (0 sorry; axioms within {propext, Quot.sound, Classical.choice}). A structured novelty search found no occupying record. The artifact package is deposited on Zenodo. This note is a priority claim of record, not a full paper.

1 Introduction

Let $a(n)$ be the integer sequence with base values $a(0), \dots, a(8) = 1, 3, 13, 22, 44, 90, 196, 406, 852$ and the order-6 recurrence $a(n) = a(n-1) + 2a(n-2) + 2a(n-4) - a(n-5) - a(n-6)$ for $n \geq 9$; these are exactly the base values and the recurrence signature $(1, 2, 0, 2, -1, -1)$ of OEIS entry A180970, the number of tatami tilings of a $3 \times n$ grid with monomers allowed [6]. The identification of this recurrence-defined sequence with the tatami counting sequence rests on the OEIS record and the Erickson–Ruskey literature

*Contact: via the GitHub account Aurora0134; ORCID to be added at Zenodo upload.

[3, 4, 5] and is *not* itself formalized here. Let $b(n)$ (the corner-defect companion) be the integer sequence with $b(0) := 0$ (a conventional value on which no theorem below depends), machine-enumerated definition inputs $b(1), \dots, b(9) = 2, 8, 24, 41, 85, 177, 381, 787, 1655$, and the same order-6 recurrence for $n \geq 10$.

This note records three theorems about these two recurrence-defined sequences, produced by AI4Math, a highly automated LLM-plus-kernel pipeline in which the Lean 4 kernel is the sole arbiter of correctness. The present text is a priority claim of record — a short deposit note accompanying a Zenodo artifact package; a fuller narrative paper is prepared separately through the pipeline’s paper lane. Every theorem stated below is a *complete proof* in the pipeline’s grading: it compiles with 0 **sorry**, its axioms stay within the standard whitelist, and the statement was frozen before proving.

2 Statement

The formal development defines the two sequences and the residue pattern **pat8** as follows (frozen snapshot; the definitions below are byte-identical to the frozen statement file `01-statement.lean`, sha256 59560f57...82b187f).

Listing 1: Formal definitions of **a180970**, **pat8** and **bcorner** (verbatim excerpt, lines 12–54 of `final/01-proof-complete.lean`; byte-identical to the frozen statement file).

```

1  /-- A180970 as a recurrence-defined sequence: base values a(0..8)
   and the
2  order-6 recurrence a(n) = a(n-1) + 2 a(n-2) + 2 a(n-4) - a(n-5) -
   a(n-6)
3  for n ≥ 9 (OEIS signature (1,2,0,2,-1,-1)). -/
4  def a180970 : ℕ → ℤ
5    | 0 => 1
6    | 1 => 3
7    | 2 => 13
8    | 3 => 22
9    | 4 => 44
10   | 5 => 90
11   | 6 => 196
12   | 7 => 406
13   | 8 => 852
14   | n + 9 =>
15     a180970 (n + 8) + 2 * a180970 (n + 7) + 2 * a180970 (n + 5)
16     - a180970 (n + 4) - a180970 (n + 3)
17

```

```

18 /-- Residue pattern:  $a(n) \% 8 = \text{pat8 } (n \% 4)$ . -/
19 def pat8 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
20   | 0 => 4
21   | 1 => 2
22   | 2 => 4
23   | _ => 6
24
25 /-- Corner-defect sequence  $b\_c(n)$ , recurrence-defined version:
    tatami tilings of a
26  $3 \times n$  grid with one corner cell removed. Base values  $b(1..9)$  from
    exact enumeration
27 (machine probe,  $n \leq 20$ );  $b(0) := 0$  is a conventional value (a  $3 \times 0$ 
    grid has no corner
28 cell) — no theorem below depends on it. For  $n \geq 10$  the same
    order-6 recurrence as
29 the base sequence holds:  $b(n) = b(n-1) + 2 b(n-2) + 2 b(n-4) - b(n-5) - b(n-6)$ . -/
30 def bcorner :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
31   | 0 => 0
32   | 1 => 2
33   | 2 => 8
34   | 3 => 24
35   | 4 => 41
36   | 5 => 85
37   | 6 => 177
38   | 7 => 381
39   | 8 => 787
40   | 9 => 1655
41   | n + 10 =>
42     bcorner (n + 9) + 2 * bcorner (n + 8) + 2 * bcorner (n + 6)
43     - bcorner (n + 5) - bcorner (n + 4)

```

Theorem 1 (mod-8 period 4). *For every integer $n \geq 4$, the residue $a(n) \bmod 8$ depends only on $n \bmod 4$: with $p(0) = 4$, $p(1) = 2$, $p(2) = 4$ and $p(3) = 6$, one has $a(n) \bmod 8 = p(n \bmod 4)$.*

Theorem 2 (corner-defect parity). *For every integer $n \geq 4$, $b(n)$ is odd.*

Theorem 3 (defect–base identity). *For every integer $n \geq 10$,*

$$20b(n) = 4a(n) + 33a(n-1) - 9a(n-2) + 8a(n-3) + a(n-4) - 3a(n-5).$$

3 Proof

The complete machine proofs are reproduced below verbatim from the frozen final development. Listing 2 and Listing 3 are the full proofs of Theorem 1 and Theorem 2; Listing 4 shows the statement of Theorem 3 (its 97-line proof is included in full in the artifact package, `proofs/01-proof-complete.lean`). All listings are excerpts of the single file `01-proof-complete.lean` (207 lines, sha256 `b0f4a692...4789a1`). A narrative prose proof is not the duty of this note; it belongs to the full paper lane.

Listing 2: Complete proof of Theorem 1 (verbatim excerpt, lines 56–84 of `final/01-proof-complete.lean`).

```

1  /-- **P1 (主攻)**: the mod-8 residue of A180970(n) depends only on
    n mod 4,
2  with pattern (4, 2, 4, 6) — equivalently  $a(n) \% 8 = \text{pat8 } (n \% 4)$ 
    for all  $n \geq 4$ .
3  Statement form verbatim from the r2 smoke-round kernel-accepted
    text. -/
4  theorem tatami_mod8_period4 (n r : ℕ) (hn : 4 ≤ n) (hr : n % 4 =
    r) :
5      a180970 n % 8 = pat8 r := by
6  induction n using Nat.strong_induction_on generalizing r with
7  | _ n ih =>
8      rcases lt_or_ge n 10 with h | h
9      · interval_cases n
10         all_goals norm_num at hr
11         all_goals subst r
12         all_goals norm_num [a180970, pat8]
13     · obtain ⟨j, rfl⟩ : ∃ j, n = j + 9 := ⟨n - 9, by omega⟩
14         have hj : 1 ≤ j := by omega
15         have rec9 : ∀ m : ℕ, a180970 (m + 9)
16             = a180970 (m + 8) + 2 * a180970 (m + 7) + 2 * a180970 (m
17               + 5)
18             - a180970 (m + 4) - a180970 (m + 3) := fun m => by
19                 simp only [a180970]
20             rw [rec9 j]
21             have h8 := ih (j + 8) (by omega) ((r + 3) % 4) (by omega) (
22               by omega)
23             have h7 := ih (j + 7) (by omega) ((r + 2) % 4) (by omega) (
24               by omega)
25             have h5 := ih (j + 5) (by omega) r (by omega) (by omega)
26             have h4 := ih (j + 4) (by omega) ((r + 3) % 4) (by omega) (
27               by omega)
28             have h3 := ih (j + 3) (by omega) ((r + 2) % 4) (by omega) (

```

```

25   by omega)
26   have hml := Nat.mod_lt (j + 9) (show 0 < 4 by norm_num)
27   have rl : r < 4 := by omega
28   interval_cases r <;>
29   simp only [Nat.reduceAdd, Nat.reduceMod, pat8] at h8 h7 h5
   h4 h3 ⊢ <;>
   omega

```

Listing 3: Complete proof of Theorem 2 (verbatim excerpt, lines 86–104 of `final/01-proof-complete.lean`).

```

1  /-- **P2 (伴随)**: the corner-defect sequence  $b_c(n)$  is odd for
   all  $n \geq 4$ . -/
2  theorem bcorner_odd (n : ℕ) (hn : 4 ≤ n) :
3    bcorner n % 2 = 1 := by
4    induction n using Nat.strong_induction_on with
5    | _ n ih =>
6      rcases lt_or_ge n 10 with h | h
7      · interval_cases n <;> norm_num [bcorner]
8      · obtain ⟨j, rfl⟩ : ∃ j, n = j + 10 := ⟨n - 10, by omega⟩
9        have rec10 : ∀ m : ℕ, bcorner (m + 10)
10         = bcorner (m + 9) + 2 * bcorner (m + 8) + 2 * bcorner (m
11         + 6)
12         - bcorner (m + 5) - bcorner (m + 4) := fun m => by
13           simp only [bcorner]
14           rw [rec10 j]
15           have h9 := ih (j + 9) (by omega) (by omega)
16           have h8 := ih (j + 8) (by omega) (by omega)
17           have h6 := ih (j + 6) (by omega) (by omega)
18           have h5 := ih (j + 5) (by omega) (by omega)
19           have h4 := ih (j + 4) (by omega) (by omega)
           omega

```

Listing 4: Formal statement of Theorem 3 (verbatim excerpt, lines 106–110 of `final/01-proof-complete.lean`); proof (97 lines) in artifact package `proofs/`.

```

1  /-- **P3 (伴随)**: exact defect  $\leftrightarrow$  base linear identity for  $n \geq 10$ :
2   $20 \cdot b(n) = 4a(n) + 33a(n-1) - 9a(n-2) + 8a(n-3) + a(n-4) - 3a(n-5)$ .
   -/
3  theorem bcorner_eq (n : ℕ) (hn : 10 ≤ n) :
4    20 * bcorner n = 4 * a180970 n + 33 * a180970 (n - 1) - 9 *
5    a180970 (n - 2)
6    + 8 * a180970 (n - 3) + a180970 (n - 4) - 3 * a180970 (n -
7    5) := by

```

4 Verification evidence

- Toolchain: Lean v4.34.0 (`leanprover/lean4:v4.34.0`) [1], `mathlib4` v4.34.0 (rev `5ed29652`) [2].
- Statement integrity: frozen statement `formalized/01-statement.lean`, sha256 `59560f5735f2c8b58a31d631a4ed28806ca23292d195fa098231fb04682b187f`; final development `final/01-proof-complete.lean`, sha256 `b0f4a692990680e67817e097548471`; symmetric difference after stripping proof bodies: zero (59/59 lines character-identical).
- Kernel check: strict compile exit 0 with empty diagnostics; a byte-level scan of the whole file (comments included) finds 0 occurrences of `sorry/admit`. Axiom audit as recorded in the final-audit record (`audit/final-01-audit.md`):

```
tatami_mod8_period4 : [propext, Classical.choice, Quot.sound] (exactly the whit  
bcorner_odd         : [propext, Quot.sound]                  (subset)  
bcorner_eq          : [propext, Quot.sound]                  (subset)  
sorryAx              : absent
```

- Audit chain: gate-2 well-definedness compilation plus degeneracy probing (24/24 bare-tactic probes fail, so no theorem is closable by a bare decision procedure), statement freeze with symmetric-difference comparison, gate-3 final strict compilation and axiom audit. Evidence originals are in the artifact package `audit/`.

5 Novelty evidence

The pipeline’s C9 novelty review adjudicated this result as *no occupying record found* (proposition layer). This is a statement about search evidence, not a claim to new mathematics; the object A180970 itself is an occupied OEIS record [6], and this note claims properties of the recurrence-defined sequences, not the entry. Query channels and query-string summary (full query log, including zero-hit evidence files, in `audit/c9-record.md` of the package, copied verbatim from the source records):

channel		query summary	result
OEIS layer	sequence	9 strings: corner / short-edge / long-edge defect sequences, each original and with 1 or 2 leading terms dropped	all null
OEIS layer	sequence	positive control	hits A180970
		22, 44, 90, 196, 406, 852	(channel works)
OEIS layer	record	full A180970 record re-fetched; case-insensitive scan for mod period congru	no hit
literature layer		arXiv all:"tatami" (40 results); abs:"tatami" AND abs:"congruence"; OpenAlex tatami tilings	0 relevant; no occupant
library layer		mathlib, compfiles, sequencelib, local rg for tatami / A180970	zero hits

The exclusivity re-review (2026-09-27, independent re-query before deposit) maintained the adjudication *no occupying record found*; its recorded residual gaps: CNKI not searched (login required), Semantic Scholar not re-tested, the teaching-literature layer (textbooks and problem sheets) not systematically swept, and the record covers the proposition layer only — not the combinatorial identification with tatami counting, which is layered as in Section 6.

6 Scope and limitations

Grading boundary, verbatim from the final audit: the three theorems are complete proofs, strictly as statements about the recurrence-defined integer sequences; they do not cover any combinatorial counting meaning. Not formalized here: (i) the identification $a(n) = \text{A180970}(n)$ = the tatami tiling count, which is backed by the OEIS record and the Erickson–Ruskey literature [6, 3, 4] (and the square-grid closed forms of [5] as background only); and (ii) the reading of $b(n)$ as the number of tatami tilings of a $3 \times n$ grid with one corner cell removed, which remains at the computational-conjecture layer (exact machine enumeration probe for $n \leq 20$ plus transfer-matrix general theory) and is not part of the proved content. The base values $b(1), \dots, b(9)$ are machine-enumerated definition inputs; $b(0) := 0$ is

a conventional value that no theorem depends on. This note contains no literature survey; it is a deposit record, not a survey article.

Data availability

Artifact package: Zenodo deposit (DOI to be reserved at upload; see `UPLOAD.md` in the source repository) with a public mirror on GitHub (URL recorded in `claims/tatami-mod8-defect/card.md` upon push). Note under CC BY 4.0, code under MIT.

Use of generative AI

(a) Stage mapping: candidate-proposition generation, formalization into Lean statements, proof search, and text drafting (including this note) were performed by AI (LLM) stages of the AI4Math pipeline; topic adjudication, statement-semantics verification and gate sign-off were human. (b) Model/node/budget (numbers copied verbatim from the source task ledger `tasks/20260926-beian-select/budget.log`): a single resolved model node, wire-id `anthropic/a6api-main/kimi-k3` (resolution level L2 relay), no cross-node switching; LLM sampling for the whole source task totals `llm-call 2/68` (selection smoke 1, tokens in/out 109/45; formalization roundtrip 1, tokens 821/2418); Lean compilations 22/36 (plus 22/24 on the selection smoke track). This claim-note stage used `llm-call 0` of its budget 4, and 8 LaTeX compilations of its budget 6 (counting the final rebuild that embedded this number and the author handle; ledger `claims/tatami-mod8-defect/budget.log`). (c) Final arbiter: All statements and proofs reported here were checked by the Lean 4 kernel: every proof compiles with no `sorry`, and `#print axioms` reports only {propext, Quot.sound, Classical.choice}. (d) The human author reviewed and is responsible for all content; the AI system is not an author.

Author contributions

CRedit-style: the human author was responsible for topic selection and adjudication, verification of statement semantics, gate sign-off, and the decision to deposit; the AI4Math pipeline (an LLM system) performed proposition generation, proof search, and text drafting. The AI system is not listed as an author.

References

- [1] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. *CADE-28*, LNCS 12699, pp. 625–635, 2021. https://doi.org/10.1007/978-3-030-79876-5_37
- [2] The mathlib Community. The Lean mathematical library. *CPP 2020*, pp. 367–381, 2020. <https://doi.org/10.1145/3372885.3373824>
- [3] A. Erickson, F. Ruskey, M. Schurch, and J. Woodcock. Auspicious tatami mat arrangements. *arXiv:1103.3309 [math.CO]*, 2011. <https://arxiv.org/abs/1103.3309>
- [4] A. Erickson, F. Ruskey, J. Woodcock, and M. Schurch. Monomer-dimer tatami tilings of rectangular regions. *Electron. J. Combin.* 18(1), #P109, 2011. <https://doi.org/10.37236/596>
- [5] A. Erickson and M. Schurch. Monomer-dimer tatami tilings of square regions. *arXiv:1110.5103 [math.CO]*, 2011. <https://arxiv.org/abs/1110.5103>
- [6] The OEIS Foundation. Entry A180970 (F. Ruskey, 2010): number of tatami tilings of a $3 \times n$ grid (with monomers allowed). *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/A180970>